

The Salty Harvest: Mending Hong Kong's Cracking Rice Bowl

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A veteran rice farmer from Mekong, used to frequently check the weather to predict the outcome of crops. He tastes the water in his irrigation canals instead. If it tastes salty, that means his harvest is ruined. And the reason behind it is saltwater spreading beyond its original boundary, which has turned his farm into a graveyard of withered rice. Given that Southeast Asia is the world's primary rice exporter, a reduced yield will inevitably restrict Hong Kong and other countries's supply and 'crack the rice bowl'."

The food system in Hong Kong is actually more vulnerable than it appears. It should be noted that Hong Kong currently imports over 90% of rice (Legislative Council of the Hong Kong Special Administrative Region, 2026). This implies that the decreasing harvest in Southeast Asia could undermine the stability of the food system in Hong Kong. Moreover, Southeast Asia is the primary exporter of rice globally (International Rice Research Institute, 2022); Therefore, this is only a regional issue, but a global concern. An agriculture report indicates that climate change is reducing the regional rice yields by 20% by 2050 (PIB Delhi, 2024). This fact creates an important signal that action must be taken to adapt to the change, transforming our diets and the desire to preserve personal choice. We cannot wait until the market fails to transform our food system.

And the solution to this food crisis is not that complicated. Currently our food supply is heavily dependent on climate-vulnerable monocultures. To protect our future diet every day, we should alter our diet to include “ancient grains” like millet and sorghum, since these climate-resilient alternative grains can grow healthily under extreme heat, high-saline conditions and which devastate farms in the Southeast-Asia region (Ghatak et al., 2021). Utilizing these grains as climate-smart crops as a replacement of traditional rice ensures Hong Kong and other countries maintain their culinary habits while the rice supply is under high-risk monocultures.

However, it is impossible to force citizens to change their dietary habits in a short period of time. To promote sustainable dietary habits, the government should employ “Choice Architecture”. According to Meier et al. (2022), individuals often stick with the default option because they are perceived as the “endorsed” choice with least cognitive effort. By slowly implementing the integrating climate-resilient grains into public food procurement such as school and hospital catering, the government can use this for enacting environmental policy, and these initiatives shift public consumption habits toward sustainability smoothly.

Lastly, the transition in diet habits of the public will be more effective and stable with the aid of local supply chain innovation. Apply controlled-environment agriculture and growing climate-resilient varieties like millets within the farm like Northern Metropolis in Hong Kong helps reduce reliance on import food. The localized food-supply chain provides a steady supply of grains to meet the huge demand of rice regionally. This multifaceted strategy, including the modification of grains, optimization of choice architecture, and localization of farming technology, allow the city to protect and build a more secured rice supply chain and prevent the city from being a passive victim in the food system due to climate change.

All these actions should be taken to shift the food system towards climate-smartness, delivering several benefits to cities, farmers and individuals. For cities, these initiatives decouple its food security from the climate impact of Southeast Asian monoculture, making the city less vulnerable to broken supply chains. For farmers, a stabilized supply of salt-tolerant grains offer an alternative product for them to export, allowing them to survive the farming crisis caused by climate change.

Beyond environmental resilience, these “ancient grains” like millet offer significant health advantages with more nutrients and fiber (Borah et al., 2023). While the policy maker and producer take action, individuals should replace white rice with these healthier climate-resilient alternatives in meals, since it creates the market demand. This small but important effort smooths the path for government policy. Ultimately, this collective shift can save the “rice bowl” of many countries, even when climate instability is destroying the production of traditional crop yields.

The future of the food system requires everyone’s action to secure. Aligning government policy with choice architecture with localized cultivation and individual habit shifts, a more resilient food system will be built and the proactive adaptation will ensure our “rice bowl” remains full and healthy in an era of climate instability.

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